CLAE: A Compact 12.5 Hz Bengali Speech Representation Encoder

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Abstract

We study a compact general-purpose representation encoder for 16 kHz Bengali speech. Its 15.29M-parameter frontend and encoder produce 256-dimensional features z at 12.5 frames/s for downstream frozen-feature discrimination tasks; the full training model has 23.80M parameters. The 12.5 Hz resolution aligns with low-frame-rate neural audio codecs and tokenizers such as Mimi, DualCodec, and NanoCodec (Défossez et al., 2024; Li et al., 2025; Casanova et al., 2025), while CLAE remains continuous: its floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not bitrate-comparable. A separate per-frame projector maps z to a 64-dimensional space p. Training uses waveform reconstruction as an auxiliary self-supervised pretext and information-retention objective, non-contrastive global/local view consistency in p to encourage nuisance-invariant features, and Variance-Invariance-Sketching Regularization (VISReg) in p to support non-collapsed, well-distributed representations. At the selected 210k-step checkpoint, frozen z features support the reported emotion, age, gender, and speaker probes; reconstruction on 400 held-out Common Voice Bengali clips gives MR-STFT 0.747, STOI 0.850, ESTOI 0.762, and PESQ-WB 2.146. A matched 25k-step study shows that removing reconstruction worsens reconstruction metrics and two temporal-emotion probes in this setting. These results are single-seed point estimates and objective proxy measurements; they do not establish encoder fine-tuning performance, statistical significance, subjective audio quality, or broad task coverage.

1 Introduction

Compact speech encoders are useful when their frozen features retain information for several downstream discrimination tasks without requiring task-specific encoder adaptation. This frozen-feature framing follows speech representation work such as wav2vec 2.0 and HuBERT (Baevski et al., 2020; Hsu et al., 2021) and the lightweight-head evaluation protocol of SUPERB (Yang et al., 2021). Modern neural audio codecs and tokenizers also commonly operate at low temporal resolutions, making 12.5 Hz a practical design point for a compact speech representation; DualCodec and NanoCodec, for example, report 12.5 Hz codec configurations (Li et al., 2025; Casanova et al., 2025). This does not make the systems bitrate- or architecture-equivalent. CLAE is continuous rather than a codec: its floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not directly comparable to codec bitrate.

We present CLAE, a 23.80M-parameter continuous-latent training model trained on an approximately 2,000-hour inventory assembled from four Bengali speech sources. Its 15.29M-parameter frontend and encoder produce a 256-dimensional frame-level representation z at 12.5 Hz; downstream probes keep this feature extractor frozen and consume z. A distinct 64-dimensional projector output p is used only by two representation objectives: direct global/local view consistency and Variance-Invariance-Sketching Regularization (VISReg; Wu et al., 2026). View consistency encourages features stable to the augmentation and masking nuisances used to construct the views. VISReg supports a centered, scaled, and distributed projector population, countering degenerate representations. Reconstruction is an auxiliary self-supervised pretext and information-retention objective: the decoder generates a waveform from a corrupted copy of z, and the generated waveform is compared with the clean source in an 80-bin log-mel domain.

The contribution is empirical and implementation-specific. We provide: (i) a compact continuous 12.5 Hz Bengali speech representation encoder evaluated through frozen-feature discrimination probes; (ii) a training objective that combines reconstruction for self-supervised information retention, non-contrastive view consistency for nuisance invariance, and frame-level VISReg for anti-collapse and distribution regularization; (iii) a Conformer-derived encoder with a project adaptation of Manifold-Constrained Hyper-Connections (mHC); and (iv) completed evaluations of the selected 210k-step checkpoint together with a matched 25k-step ablation suite. The completed probes make CLAE a candidate general-purpose Bengali speech representation for the stated tasks, not a demonstrated foundation model: we do not claim encoder fine-tuning performance, novelty of the individual components, deployment-codec capability, or broad language or population coverage.

2 Related Work

Joint-embedding learning. Image JEPA introduced representation-space prediction with distinct context and target pathways (Assran et al., 2023); A-JEPA adapted masked-target JEPA ideas to audio (Fei et al., 2023). LeJEPA instead studies direct view agreement with isotropic-Gaussian regularization (Balestriero and LeCun, 2025). CLAE uses a JEPA-inspired, non-contrastive per-frame view-agreement loss, but it is not canonical predictive JEPA: it has one shared encoder and no target encoder, exponential moving average, learned predictor, stop-gradient target path, masked-region prediction, or negative pairs. It also differs from LeJEPA in domain, waveform view construction, grouped consistency loss, frame-level population, supporting regularizer, and reconstruction.

Distribution regularization. VICReg combines invariance, per-dimension variance control, and covariance regularization (Bardes et al., 2022). VISReg retains explicit center and scale terms but replaces covariance control with random one-dimensional projections whose sorted samples are matched to Gaussian quantiles (Wu et al., 2026). We use the exact name Variance-Invariance-Sketching Regularization as a supporting anti-collapse and distribution constraint, not as the central representation-learning premise. The project adopted the recent VISReg preprint in July 2026 and applies a local, frame-level adaptation described in Section 3.3; no controlled VISReg-versus-SIGReg experiment is available.

Continuous and discrete audio representations. Continuous Audio Language Models use a continuous variational bottleneck, KL regularization, additional speech distillation, and generative next-frame modeling (Rouard et al., 2025); those components are not used here. UniWav studies joint representation and generation objectives for speech (Liu et al., 2025a). By contrast, UniAudio and UniTok-Audio illustrate autoregressive generation over discrete audio tokens (Yang et al., 2024; Liu et al., 2025b). Neural codecs such as SoundStream and EnCodec learn quantized, bitrate-controlled representations (Zeghidour et al., 2021; Défossez et al., 2023). Mimi, introduced with Moshi, is a separately pretrained common-evaluation reconstruction reference in our 25k study (Défossez et al., 2024). CLAE's continuous floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not directly comparable to codec bitrate.

Sequence encoder and residual connections. The encoder builds on the Conformer macaron block (Gulati et al., 2020). Its local class name refers to FastConformer, but it does not implement the published architecture's complete downsampling and scalable/local-attention design (Rekesh et al., 2023); we therefore call it a Conformer-derived encoder. Selected layers use a project adaptation of Manifold-Constrained Hyper-Connections (Xie et al., 2025), not a codebook, clustering method, or quantizer.

3 Method

3.1 Architecture

For a 16 kHz mono waveform x, the convolutional frontend has five Conv1D stages with channels $[128, 256, 384, 512, 512]$, kernels $[10, 8, 8, 4, 4]$, and strides $[5, 4, 4, 4, 4]$, with GroupNorm and GELU. Its total stride is 1,280 samples, yielding approximately 12.5 frames/s. An eight-layer, 256-dimensional Conformer-derived encoder uses eight attention heads, a 1,024-dimensional feed-forward sublayer, rotary full self-attention, macaron feed-forward modules, a 9-tap convolution branch, dropout 0.1, and squeeze-and-excitation. It produces

$$z = E(F(x)), \qquad z \in \mathbb{R}^{T' \times 256}. \tag{1}$$

A per-frame MLP projector with dimensions $256 \to 512 \to 64$ produces $p = P(z)$. The decoder and downstream frozen-feature probes use

z, whereas the consistency and VISReg objectives use p.

The FiLM-conditioned waveform decoder starts with 768 channels, reverses the frontend with strides $[4, 4, 4, 4, 5]$, and uses two residual blocks per upsampling stage with dilations $[1, 3, 9]$. The full training model has 23,803,125 parameters. The frontend, encoder, projector, and decoder contain 2,890,240, 12,403,220, 165,440, and 8,344,225 parameters, respectively. The frozen frontend–encoder feature extractor therefore contains 15,293,460 parameters.

3.2 Waveform views and consistency objective

Training first samples one three-second source segment, and every view is derived from that same pre-cropped waveform. Two global views receive independently sampled waveform augmentation. Four local views remain full-length three-second waveforms but receive stronger waveform augmentation and waveform-aligned span masking; their frontend features can receive an additional independent frame mask. The waveform augmentations are gain, additive noise, centered low-pass filtering, and clipping only: they do not shift, speed-perturb, resample, independently crop, or temporally warp the signal. Waveform masks are constructed on the 1,280-sample frontend-frame grid and expanded onto the corresponding waveform samples; the optional frontend mask is also defined on that frame grid. Thus, frame index t refers to the same time interval across views, although its waveform content may be augmented or masked. All six views pass through the same frontend, encoder, and projector.

Let p_g and p_l denote projector features grouped into global and local views. For each example and frame, the center $\bar{p}_g$ is the mean of the two global projector features. The implemented consistency loss is

$$\mathcal{L}_{\text{view}} = \text{mean}\big[(p_g - \bar{p}_g)^2\big] + \text{mean}\big[(p_l - \bar{p}_g)^2\big] . \quad (2)$$

The two group means are added, so global and local groups have equal aggregate contribution despite containing two and four views. This JEPA-inspired objective is non-contrastive per-frame global/local agreement, not canonical context-to-target or masked-region prediction, and it is not a direct utterance-identity or class-discrimination loss. It encourages invariance to the specified augmentation and masking nuisances, which can be useful for the downstream discrimination probes, but those probes are separate evaluations of z.

3.3 VISReg adaptation

The supporting VISReg implementation flattens projector frames from all views and examples into one population; it does not pool them. Under distributed training, the population is gathered across workers. Its effective shape is $N = 1$, $B = T'VB_{\text{global}}$, and $D = 64$. For population points $q_i \in \mathbb{R}^D$, it penalizes the population mean, centers the points, computes a per-dimension RMS scale with division by $\sqrt{B}$, and penalizes deviation from unit scale. Normalization uses a detached scale. On every forward pass, 256 normalized Gaussian directions are sampled. Values projected onto each direction are sorted and matched to standard-normal quantiles $\Phi^{-1}(i/(B+1))$. Center, scale, and projected-distribution shape terms have equal internal weight.

This implements the core VISReg center/scale/shape construction with project-specific frame-level and distributed adaptations. The version used by the 210k checkpoint is the repository implementation at commit `35be6652f7a2`. Relative to the later upstream implementation, it gathers across workers, adds 10^{-6} to scale rather than using a lower clamp, fixes equal internal weights, and casts cached target quantiles to the activation dtype. The main run uses BF16. We do not describe this version as identical to the current upstream implementation.

3.4 Reconstruction and total objective

The decoder receives z from the first global view. Before decoding, an independent span mask removes 50% of frames in spans of 4–24 frames and Gaussian noise with standard deviation 0.03 is added. The target remains the clean source waveform. The decoder outputs a waveform, which is compared with the clean waveform using an 80-bin log-mel loss with FFT size 1,024, window length 1,024, hop length 256, a Hann window, magnitude weight 1.0, log-magnitude weight 1.0, and spectral-convergence weight 0. Multi-resolution STFT code exists but is not an additional training objective for the selected run.

The selected objective is

$$\mathcal{L}_{\text{total}} = 1.0\mathcal{L}_{\text{mel}} + 0.3\mathcal{L}_{\text{view}} + 0.7\mathcal{L}_{\text{VISReg}}. \quad (3)$$

These values are coefficients for losses with different scales; 0.7 is not a percentage of the total

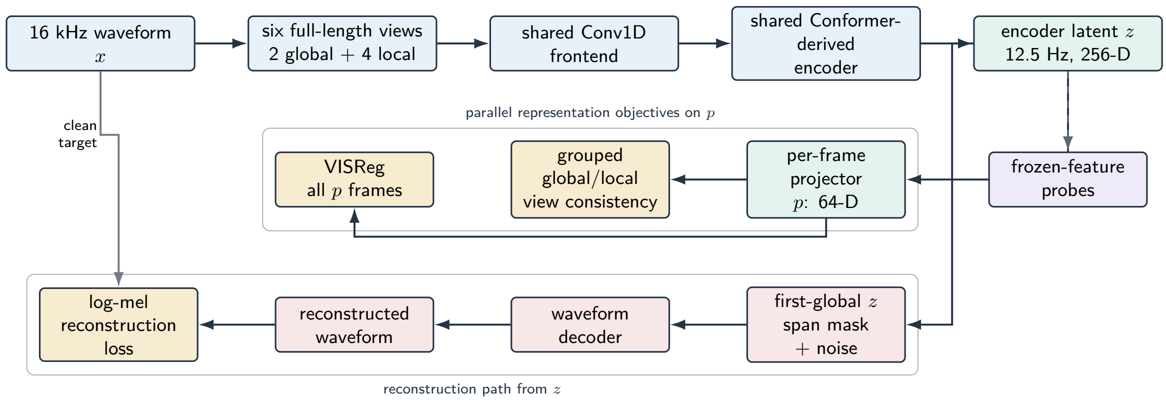

Figure 1: Implemented training and evaluation paths. Two global and four full-length local views of the same pre-cropped segment share the frontend, encoder, and projector. Local views use stronger timing-preserving augmentation and frame-aligned masking, not shorter temporal crops. The JEPA-inspired non-contrastive per-frame agreement and supporting VISReg constraint act on projector outputs p; decoder corruption and waveform reconstruction use the first-global encoder latent z, and frozen-feature probes consume z.

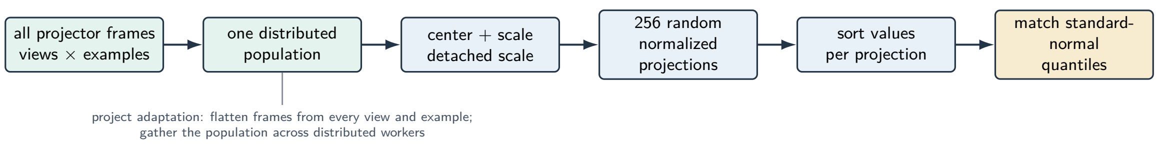

Figure 2: Supporting project-specific VISReg constraint. Projector frames from every view and example are gathered into one distributed population, centered and scaled, projected onto 256 freshly sampled normalized directions, sorted, and matched to standard-normal quantiles.

objective. Vector quantization, a variational bottleneck, KL divergence, adversarial loss, feature matching, and autoregressive prediction are inactive in the selected run.

3.5 mHC adaptation

At zero-indexed encoder layers 2 and 5, mHC widens the residual path to two streams. It applies learned static pre/post mixing and a Sinkhorn-constrained residual matrix with identity blending, uses ten Sinkhorn iterations, wraps complete encoder layers, and returns to one stream by a project-specific uniform mean readout. These choices adapt the reference mHC design rather than reproduce its complete large-model implementation. Section 5.2 reports mixed, probe-dependent evidence for this component.

4 Data and Training

The training manifests combine four Bengali speech sources, with 1,310,076 utterances and close to 2,000 total hours of audio: Common Voice (Ardila et al., 2020), OpenSLR-53 (Kjartansson et al., 2018), RegSpeech12 (Hassan et al., 2025), and Shrutilipi (Bhogale et al., 2023). Table 1 gives the source inventory. The manifests use a 95/5 stratified train/validation split with seed 42; Table 2 gives the resulting counts. No exact audited unique duration is available. Common Voice and OpenSLR-53 both contribute to pretraining, so evaluations on those sources are in-domain frozen-feature evaluations with unassessed train–evaluation overlap. The inventory does not establish demographic or dialect balance, and no deduplication or train–evaluation leakage audit was performed. SUBESCO (Sultana et al., 2021), used for emotion evaluation, is absent from the documented training inventory.

The selected checkpoint comes from run `large-2kh-packed-210k-tail-lr-1e4`.[1] Training was deliberately stopped at 210,000 steps,

[1] Repository-relative checkpoint path: `runs/large-2kh-packed-210k-tail-lr-1e4/checkpoints/last.pt`.

Dataset	Utterances
Common Voice Bengali	1,052,178
OpenSLR-53	218,703
regspeech12	21,313
shrutilipi	17,882
Total	**1,310,076**

Table 1: Bengali speech sources in the training manifests.

Split	Utterances
Train	1,244,572
Validation	65,504

Table 2: Stratified 95/5 manifest split (seed 42).

and the resulting terminal checkpoint is reported; it was not selected from a downstream-metric checkpoint sweep. Training uses BF16, AdamW, batch size 42, gradient accumulation over four batches, and gradient clipping at 1.0. The complete 210k trajectory is distributed across four W&B records: `20260717_124435`, `large-2kh-packed-recovery-lr-half`, `large-2kh-packed-300k-tail-lr-1e4`, and `large-2kh-packed-210k-tail-lr-1e4`. Each record resumes the same model trajectory, although checkpoint directories and run identifiers change between machines. The first continuation resumed the 45k checkpoint under the packed base configuration; the 50k continuation reduced the base learning rate to 5×10^{-4}; and the later continuations used 10^{-4} with a 300k scheduler horizon before the final maximum was set to 210k. The scheduler is reconstructed from the completed step and active resume configuration rather than restored as independent state, so the trajectory is continuous in checkpoint lineage but not one uninterrupted default cosine schedule.

The effective batch is 168 source segments per optimizer step. At 210,000 steps this corresponds to 35.28M sampled source segments, or 105.84M seconds (29,400 hours) of nominal source-audio exposure. It is also about 28.35 training-manifest-equivalent sampled utterance counts. These are exposure-equivalent quantities, not unique examples or duration: augmentation, random cropping, packed-data shuffling, and repeated sampling alter what is observed. The six representation views increase computation but are not six independently sampled source recordings.

There is no active in-loop validation. Configuration fields for a validation manifest, evaluation interval, and validation batches do not trigger validation in the training path. W&B records training provenance and diagnostics; its curves are not validation curves and are not used here as evidence of generalization.

Figure 3 shows the four W&B records for this single 210k trajectory in separate panels. The first record contains both original and resumed logging and overlaps the 50k recovery record from step 50,050 through 59,250 because that continuation resumed from the 50k checkpoint. The third segment completed successfully through step 170,700; its W&B terminal-state flag is inaccurate. The final record continues the same training trajectory from the renamed `runs/bengali-210k/checkpoints/last.pt` checkpoint and logs steps 170,750 through 210,000. We keep the panels separate to expose the overlap, resumptions, and manual schedule changes rather than hiding them in a stitched curve.

5 Evaluation

We report only completed artifacts that identify the checkpoint, data, protocol, and metric. The selected 210k checkpoint is evaluated separately from the matched 25k ablations; their absolute scores should not be interpreted as a checkpoint-matched comparison. Reconstruction metrics are objective proxies. PESQ (Rix et al., 2001), STOI (Taal et al., 2010), ESTOI (Jensen and Taal, 2016), and MR-STFT (Yamamoto et al., 2020) do not substitute for listener judgments.

Where available, we compare frozen CLAE features with WavLM (Chen et al., 2022), Whisper (Radford et al., 2023), ECAPA-TDNN (Desplanques et al., 2020), emotion2vec (Ma et al., 2024), Mimi (Défossez et al., 2024), and the exact evaluated Higgs tokenizer, `bosonai/higgs-audio-v2-tokenizer` at revision `403fbacf2f60caaa102f893fdfabb694619b2417` (Boson AI, 2025). These are purpose-built systems with different training objectives and specializations; they are reported as task-specific references rather than a homogeneous family of representation models.

5.1 Selected 210k checkpoint

Table 3 reports the selected completed metrics used for the main analysis. Reconstruction uses a fixed order of 400 three-second Common Voice Ben-

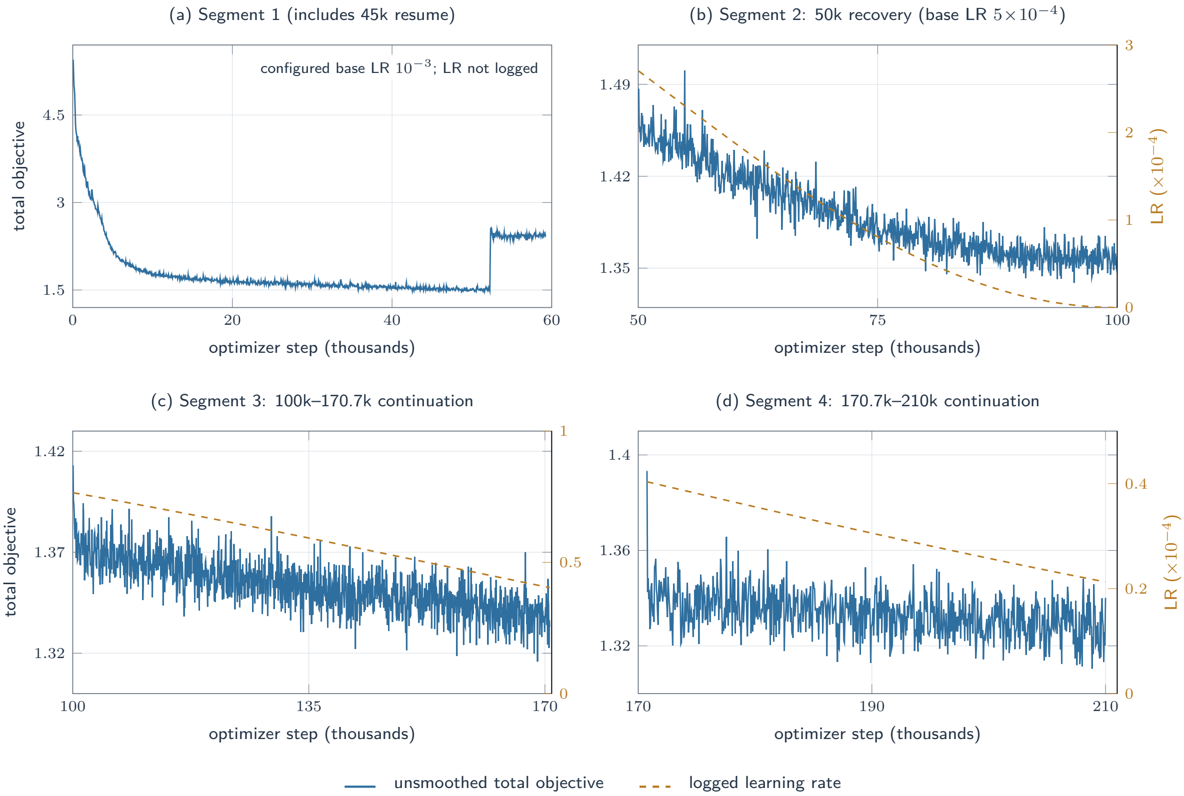

Figure 3: The complete 210k training trajectory across four W&B records and three resume boundaries. Curves are unsmoothed and panels remain separate to show the overlapping 50k recovery and manual learning-rate changes; objective-axis ranges differ by panel. The first record has no logged learning-rate series. These are training diagnostics, not validation or convergence evidence.

gali validation clips and FP32 inference. Classification probes keep the frontend and encoder frozen. Emotion, age, and gender use speaker-disjoint folds. Closed-set speaker identification holds out utterances from the same enrolled speakers. Speaker verification uses cosine-scored all-pairs trials, which is a diagnostic rather than a standard enrollment/test benchmark. To keep the reporting consistent with the compact evaluation setting, Table 3 gives the fixed seed-0 point estimate for learned probes; reconstruction and verification have no probe seed.

The table establishes performance only under the stated protocols. Common Voice results are in-domain, and train–evaluation overlap has not been audited. The speaker-identification result is closed-set and does not imply open-set verification performance. The age and gender probes measure label prediction under their speaker-disjoint splits; they do not evaluate fairness or population coverage.

5.2 Matched 25k ablations

The ablation evidence comes only from the allowed `clae-full-evaluations3` artifacts. All model conditions stop at 25,000 steps while retaining the base configuration's 100,000-step learning-rate horizon. Available reconstruction aggregates use the same fixed 400 Common Voice Bengali clips in FP32. The temporal-attention probe uses 1,708 training and 392 held-out SUBESCO utterances. The temporal Transformer is a different probe family, not another seed of the attention probe. Each temporal result is a single-seed point estimate.

At this budget, removing reconstruction substantially worsens all four reported reconstruction metrics and both temporal-emotion probe results. This shows the effect of removing the training signal in this setting; it does not establish that z collapsed or that reconstruction is universally necessary. The 25 Hz condition improves STOI, ESTOI, PESQ-WB, and temporal-Transformer F1 relative to the 12.5 Hz base, while doubling the latent frame rate. It is therefore not a matched representation-rate or bitrate comparison.

Task	Data and protocol	Metric	Result
Reconstruction	Common Voice Bengali, fixed 400 clips	MR-STFT ↓	0.747
		STOI ↑ / ESTOI ↑	0.850 / 0.762
		PESQ-WB ↑	2.146
Emotion (linear)	SUBESCO, 5 speaker-group folds, 7,000 utt.	macro-F1 ↑	0.482
Emotion (temp. attention)	SUBESCO, 1,708 train / 392 held-out utt.	macro-F1 ↑	0.359
Emotion (temp. Transformer)	SUBESCO, 4 speaker-group folds, 7,000 utt.	macro-F1 ↑	0.516
Age	Common Voice, 5 speaker-group folds, 15,475 utt.	balanced acc. / macro-F1 ↑	0.259 / 0.216
Gender	Common Voice, 5 speaker-group folds, 15,690 utt.	balanced acc. / macro-F1 ↑	0.954 / 0.928
Closed-set speaker ID	167 speakers, 666 train / 334 test utt.	accuracy ↑	0.656
Speaker verification	334 speakers, 2,000 utt., all-pairs, mean+std	EER / minDCF ↓	0.276 / 0.988

Table 3: Completed evaluations for the selected 210k checkpoint. "Temp." denotes temporal attention or the temporal Transformer, respectively. Probe values are fixed seed-0 point estimates. The tasks use different datasets and protocols and should not be combined into an overall ranking.

Condition	MR-STFT ↓	STOI ↑	ESTOI ↑	PESQ-WB ↑	Attn. F1 ↑	Transf. F1 ↑
Packed base, 25k	0.850	0.828	0.715	1.970	0.341	0.441
Representation-only, 25k	5.652	0.319	-0.001	1.103	0.132	0.197
No mHC, 25k	0.829	0.824	0.721	1.977	0.289	0.449
25 Hz, 25k	0.838	0.857	0.767	2.090	0.318	0.458
No decoder corruption, 25k	—	—	—	—	0.311	0.450
Mimi, 8 quantizers (nominal 1.1 kbps)	0.869	0.829	0.707	2.007	n/a	n/a

Table 4: Matched 25k-step ablations and the Mimi common-evaluation reconstruction reference. "Attn." denotes the temporal-attention probe and "Transf." the temporal-Transformer probe. A dash means that no completed artifact is available. Mimi is separately pretrained and evaluated here only on the common reconstruction inputs.

The mHC result is probe-dependent. Removing mHC slightly lowers MR-STFT and raises temporal-Transformer F1, whereas retaining mHC raises temporal-attention F1. The available evidence does not support a uniform benefit. Packed base and the separately pretrained Mimi reference exhibit a metric-dependent trade-off on common evaluation inputs: packed base is lower on MR-STFT and higher on ESTOI; Mimi is marginally higher on STOI and PESQ-WB. Neither dominates across the reported metrics. Reconstruction-only is absent from this older common-400 artifact, and no completed reconstruction metrics are available for the no-decoder-corruption condition. A separate reduced fixed-subset FP32 comparison includes reconstruction-only (Section B) but uses different sample budgets and must not be merged with these rows.

6 Limitations

There is no active in-loop validation. The 210k results are limited to completed existing artifacts and the reported probe values are fixed-seed point estimates; the 25k temporal ablations also use one probe seed. We report no paired-bootstrap analysis, confidence intervals, statistical significance, stability, or robustness claims. No human listening study was conducted, so reconstruction evidence is limited to objective proxy metrics. There is no controlled VISReg-versus-SIGReg comparison.

The training inventory was not audited for demographic or dialect coverage, deduplication, or leakage. Common Voice and OpenSLR evaluations are in-domain because both sources contribute to pretraining, and their train–evaluation overlap is unknown. The available evaluation artifact does not preserve a paper-ready age-label ontology or trivial baseline, which limits interpretation of the age probe. The all-pairs speaker-verification protocol is not a standard enrollment/test benchmark. We draw no conclusion about linguistic retention. The 25 Hz ablation changes frame rate, and mHC effects are mixed. VISReg and mHC are recent preprints used through project-specific adaptations; the reported findings concern the exact local implementation and configuration.

7 Conclusion

CLAE is a compact 12.5 Hz continuous Bengali speech representation encoder trained with reconstruction as an auxiliary self-supervised information-retention objective, non-contrastive global/local view consistency for nuisance invariance, and frame-level VISReg for anti-collapse and distribution regularization. Under the reported protocols, its frozen features support the selected downstream discrimination tasks; this is evidence for a candidate general-purpose Bengali speech representation within the evaluated scope, not for encoder fine-tuning or a broad foundation-model claim. The matched 25k study shows a setting-specific benefit from retaining reconstruction, a frame-rate trade-off at 25 Hz, mixed mHC effects, and a metric-dependent comparison with Mimi. CLAE remains continuous and is not bitrate-comparable to quantized codecs. These conclusions do not extend beyond the completed tasks and artifacts.

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A Complete Published Evaluation Tables

The tables below retain completed scalar results from the audited `clae-full-evaluations2` and `clae-full-evaluations3` artifacts. Missing directories, skipped tasks, failed evaluations without result files, and support-only caches are not converted into results. A reported $\pm$ value is the sample standard deviation across saved stochastic probe seeds, not a confidence interval. The deterministic linear probes produced identical scalar values across their nominal seeds and are therefore shown as point estimates.

A.1 Evaluation implementation and reproducibility

The selected model is the terminal step-210,000 checkpoint `runs/large-2kh-packed-210k-tail-lr-1e4/checkpoints/last.pt`. CLAE probes use only the frozen final frontend–encoder output z (256 dimensions at 12.5 Hz): they neither combine encoder layers nor use the projector p. Evaluation audio is mixed to mono and resampled to 16 kHz. For CLAE feature extraction, a clip is split into non-overlapping segment-length windows, the final window is right-padded, and frame features are concatenated before pooling; this avoids applying the globally attentive encoder to an out-of-distribution long window. Utterance probes use mean+standard-deviation pooling unless noted; speaker ID uses mean pooling, and verification evaluates both mean and mean+standard-deviation pooling. Pooling excludes known right-padding frames when a duration is available.

External representations are frozen and pinned to revisions: WavLM-base-plus (`4c66d4806a428f2e922ccfa1a962776e232d487b`) and Whisper-tiny (`169d4a4341b33bc18d8881c4b69c2e104e1cc0af`) use final hidden states; WavLM is final-layer only, not a learned weighted layer aggregation. ECAPA (`0f99f2d0ebe89ac095bcc5903c4dd8f72b367286`) uses its utterance embedding; emotion2vec (`0c9a3152734f9d7a7a05b4ee6bfb3c109d288664`) uses continuous FunASR frame features; Mimi (`89091b3e466eb6a9d11e537bf26b144f194978f7`) uses its continuous pre-quantization latent; and Higgs Audio V2 uses `bosonai/higgs-audio-v2-tokenizer` at revision `403fbacf2f60caaa102f893fdfabb694619b2417` and decodes its quantizer output to continuous vectors. No codec probe substitutes integer code IDs for continuous features. The separate Mimi reconstruction comparator fixes eight quantizers (nominal 1.1 kbps), rather than reusing the pre-quantization probe feature.

The 210k run directory, including its referenced `config.yaml`, is not present in this project snapshot. Frozen CLAE feature probes remain reconstructible from the checkpoint-embedded `cfg`; by contrast, the reconstruction and attention-ASR entry points take a separate configuration path, so a literal rerun of those commands also requires that missing configuration artifact.

A.2 Matched 25k ablation coverage

Table 4 reports every completed scalar result from the matched 25k common-400 artifact set. The suite was intended to contain six model conditions, but its reconstruction-only upload is absent. The no-decoder-corruption condition has completed temporal-emotion probes, while its reconstruction task recorded failure without a result file. Ordinary ASR, emotion, and gender probes were explicitly skipped throughout this older ablation set. These statuses explain the dashes in Table 4; they are not zero scores; they do not imply that reconstruction-only was never evaluated under a separate protocol.

B Reduced Fixed-Subset FP32 Evaluation

This study is separate from the preceding full-data evaluation and does not replace or extend its rows. It uses one fixed seed (data, split, and probe seed 0 where applicable) and FP32 CLAE inference with CUDA autocast disabled. Reconstruction uses 100 three-second clips; generic and temporal emotion use the same 1,400 SUBESCO clips, with 20 training epochs for the temporal probes; age and gender each use 2,000 Common Voice clips; speaker identification uses 500 OpenSLR-53 clips; and verification uses 800 OpenSLR-53 clips with 20,000 fixed trials. ASR was deliberately not evaluated. All values are descriptive one-seed outcomes without confidence intervals, error bars, significance tests, or claims of robustness.

Packed 210k has the lowest STFT loss, while the 25k no-decoder-corruption condition has the strongest PESQ, STOI, age, speaker-identification, and both temporal-emotion results. In the direct reduced comparison, reconstruction-only leads generic emotion descriptively (43.89 vs. 41.59 macro-F1), whereas packed base is slightly higher on age (26.11 vs. 25.89) and the two temporal probes (35.21 vs. 34.97 and 42.30 vs. 41.13). These one-seed differences are small and mixed, so they do not demonstrate a uniform benefit of the view-consistency-plus-VISReg branch. The 25 Hz condition leads gender and verification. Representation-only is a negative reconstruction control and also trails the other trained CLAE conditions on every representation task in this reduced protocol.

Whisper-tiny leads generic emotion, Mimi leads age, and ECAPA leads gender, closed-set speaker identification, and verification. WavLM has the strongest configured temporal-Transformer baseline. The pinned eight-quantizer Mimi common-

Evaluation	Split and pooling	Head / score	Fixed full-suite protocol
Reconstruction	Fixed manifest order; valid unpadded prefix	FP32 encoder–decoder; MR-STFT, waveform L1, STOI, ESTOI, PESQ-WB	8 clips/batch $\times$ 50 batches; 3-s clips; shared four-resolution STFT contract.
Emotion, age, gender	5 speaker-group folds; mean+std	Standardized logistic regression ($C = 1$, 3,000 iterations); age uses balanced class weights	All available SUBESCO clips for emotion; all available labelled Common Voice clips for age and gender; data/split/probe seeds 0.
Speaker ID	Per-speaker held-out utterances; mean	Standardized multinomial logistic regression ($C = 1$, 3,000 iterations)	Up to 1,000 OpenSLR-53 clips; two test utterances per speaker; seeds 0.
Verification	All upper-triangle pairs; mean and mean+std	Cosine score, EER and minDCF ($P_{\mathrm{target}} = 0.01$)	Up to 2,000 OpenSLR-53 clips; no trial cap; data/split seeds 0.
Attention ASR	Frozen frame sequence; no utterance pool	2-layer, 4-head attention decoder ($d = 256$, FFN 1,024)	Up to 10,000 train and dev items; 8,000 updates, batch 16; each input is duration-validated at 15 s, encoded in 3-s padded windows and concatenated; seed 0.
Temporal emotion	Frozen frame sequence, capped at 300 frames	Attentive statistics head (128 hidden units; batch 64) or 4-fold Transformer classifier	Attention: fixed held-out speakers F_09, F_10, M_09, M_10, 40 epochs. Transformer: speaker-group folds, 30 epochs. Both use seed 0 when enabled.

Table 5: Implemented evaluation protocol. The full-suite launcher defaults to one seed (0), its original full-data limits, and no temporal-emotion tasks; temporal tasks require explicit selection. The separate reduced-suite results below use their stated smaller fixed subsets.

25k condition	Reconstruction	Temporal probes
Packed base	complete	complete
Representation only	complete	complete
No mHC	complete	complete
25 Hz	complete	complete
No decoder corruption	no result file	complete
Reconstruction only	absent	absent

Table 6: Artifact coverage for the older matched 25k common-400 ablation suite. "Complete" means that the corresponding result artifact exists in that suite; the separate reduced fixed-subset FP32 suite has its own coverage and rows.

evaluation reconstruction reference obtains STFT 0.8949 on the same 100 clips.

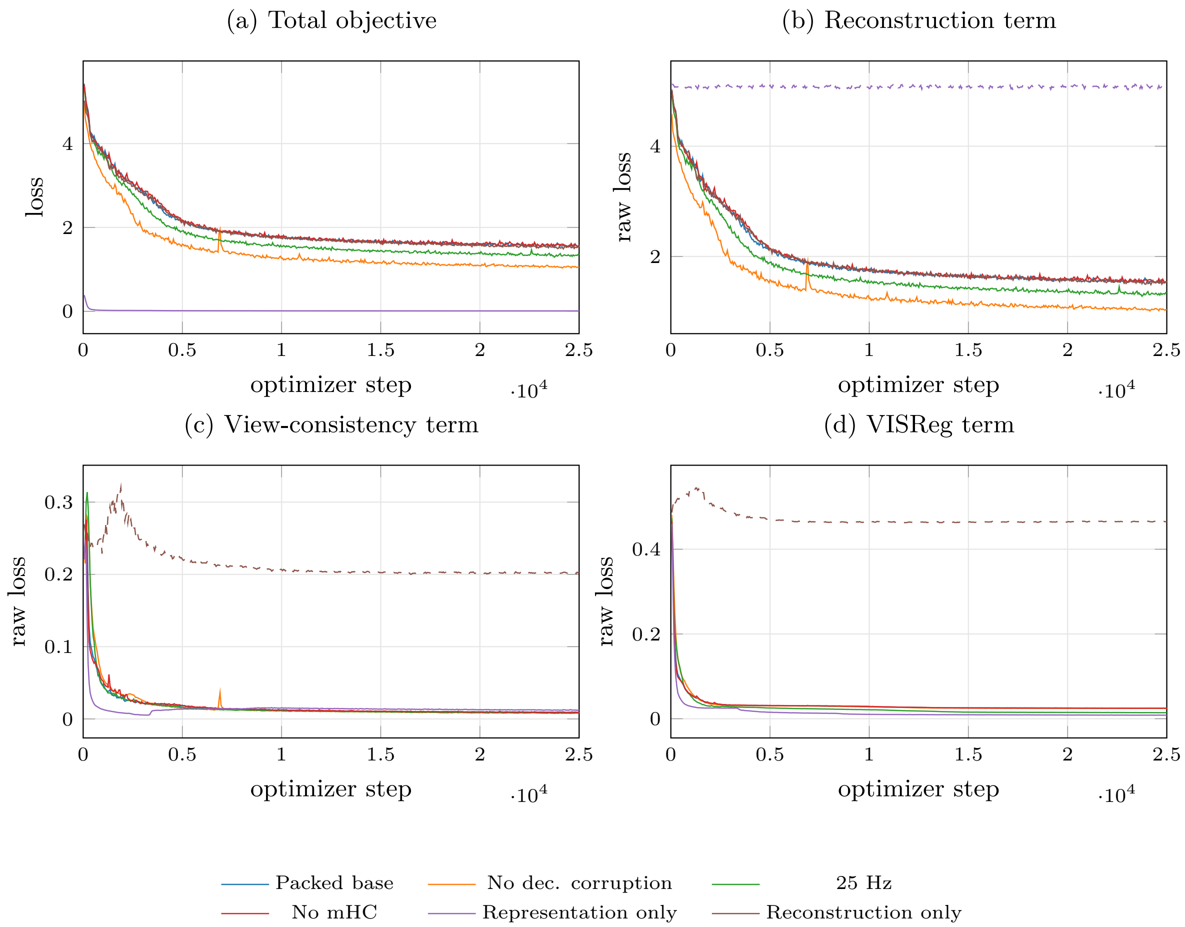

Figure 4: Unsmoothed W&B training histories for the six matched 25k conditions. Panels (b)–(d) show raw component values, not weighted contributions to the total objective. The dashed reconstruction curve for representation-only and dashed representation curves for reconstruction-only were logged diagnostically but had zero objective weight. These are training diagnostics, not validation curves or evidence of generalization.

Full-suite condition	MR-STFT ↓	CER ↓	WER ↓	Attn. F1 ↑	Transf. F1 ↑	EER ↓	minDCF ↓
No decoder corruption, 25k	0.775	0.790 ± 0.020	1.106 ± 0.027	0.301 ± 0.011	0.525 ± 0.007	0.319	0.988
Representation only, 25k	5.781	0.899 ± 0.019	1.103 ± 0.042	0.159 ± 0.024	0.227 ± 0.006	0.387	1.000
Packed selected model, 210k	0.747	0.795 ± 0.010	1.116 ± 0.014	0.341 ± 0.017	0.508 ± 0.008	0.276	0.988

Table 7: Completed full-suite reconstruction, ASR, temporal-emotion, and speaker-verification results. EER and minDCF use mean+standard-deviation pooling. These three conditions are complete in the published full-suite artifacts; the other nominal conditions have no complete full-suite result set.

Full-suite condition	Emotion acc.	Emotion F1	Age bal. acc.	Age F1	Gender bal. acc.	Gender F1	Speaker-ID acc.
No decoder corruption, 25k	0.482	0.478	0.286	0.222	0.948	0.924	0.713
Representation only, 25k	0.360	0.350	0.231	0.153	0.939	0.889	0.162
Packed selected model, 210k	0.489	0.482	0.259	0.216	0.954	0.928	0.656

Table 8: Completed deterministic linear-probe results from the same full-suite artifacts. Emotion uses SUBESCO; age and gender use Common Voice Bengali with speaker-disjoint splits; speaker identification is closed-set.

Evaluation	Features	Metric 1	Metric 2	Metric 3	Metric 4
Reconstruction	Mimi, 8 quantizers	MR-STFT 0.869	—	—	—
ASR development	WavLM	CER 0.522	WER 0.905	—	—
Temporal emotion	Random CLAE	Attention acc. 0.283	Attention F1 0.229	Transformer acc. 0.298	Transformer F1 0.284
Temporal emotion	WavLM	Attention acc. 0.413	Attention F1 0.372	Transformer acc. 0.612	Transformer F1 0.610

Table 9: Completed shared-baseline diagnostics from the published full-suite extraction. These rows use different evaluators and are grouped only to preserve the remaining completed scalar results compactly.

Features	Emotion F1 ↑	Age F1 ↑	Gender F1 ↑	Speaker-ID acc. ↑	Verification EER ↓
CLAE, selected 210k	0.482	0.216	0.928	0.656	0.276
Random CLAE	0.305	0.168	0.742	0.039	0.391
WavLM	0.602	0.249	0.936	0.296	0.339
Whisper-tiny	0.634	0.234	0.944	0.356	0.375
ECAPA	0.519	0.251	0.973	0.952	0.060
emotion2vec	0.629	0.259	0.894	0.326	0.344
Mimi	0.610	0.257	0.957	0.647	0.231
Higgs Audio V2	0.452	0.235	0.929	0.725	0.186

Table 10: External-baseline comparison accompanying the published 210k evaluation. Verification uses mean+standard-deviation pooling and the nonstandard all-pairs protocol. Models and tasks have different specializations; the columns should not be averaged into a composite ranking.

Condition	STFT ↓	PESQ ↑	STOI ↑	Emo. F1 ↑	Age F1 ↑	Gender F1 ↑	Spk-ID acc. ↑	Temp. F1 ↑	Transf. F1 ↑	EER ↓
Packed base, 25k	0.8679	1.983	0.837	41.59	26.11	93.23	67.66	35.21	42.30	27.91
Reconstruction only, 25k	0.8705	2.052	0.842	**43.89**	25.89	93.32	68.26	34.97	41.13	26.50
Representation only, 25k	4.2770	1.115	0.323	30.21	21.03	90.59	25.75	17.57	19.13	38.92
No mHC, 25k	0.8560	2.013	0.835	41.15	24.69	94.13	69.46	30.29	39.55	27.51
25 Hz, 25k	0.8419	2.139	0.865	43.74	24.77	**95.08**	70.66	32.41	43.37	**23.84**
No decoder corruption, 25k	0.7888	**2.336**	**0.882**	43.27	**27.19**	93.24	**73.65**	**36.72**	**43.56**	30.97
Packed selected model, 210k	**0.7745**	2.167	0.859	43.21	24.16	94.60	65.87	35.59	42.42	27.11

Table 11: Reduced fixed-subset FP32 CLAE evaluation. Classification scores and EER are percentages. The bold value is the best descriptive result in each column, not a statistically established winner. ASR was not evaluated.

Features	Emotion F1 ↑	Age F1 ↑	Gender F1 ↑	Spk-ID acc. ↑	Temp. F1 ↑	Transf. F1 ↑	EER ↓
Random CLAE	28.85	23.41	67.45	7.78	22.67	26.41	38.33
WavLM	53.36	25.57	94.93	34.13	31.53	**47.59**	34.19
Whisper-tiny	**59.79**	27.22	97.16	38.32	—	—	37.93
ECAPA	45.32	30.20	**99.09**	**97.01**	—	—	**6.32**
emotion2vec	55.81	25.12	85.12	36.53	—	—	35.42
Mimi	56.62	**30.39**	95.85	67.66	—	—	22.85
Higgs Audio V2	42.37	23.80	94.47	74.85	—	—	18.34

Table 12: External representations on the same reduced fixed subsets. Temporal probes were configured only for random CLAE and WavLM. Dashes mean not evaluated, not zero. Model specializations differ, so columns should not be averaged into a composite ranking.

Features	Emotion F1 ↑	Age F1 ↑
CLAE, 170k	0.496 ± 0.062	0.208
Random CLAE	0.316 ± 0.028	0.173
WavLM	0.607 ± 0.080	0.252
Whisper-tiny	0.632 ± 0.058	0.232
ECAPA	0.523 ± 0.049	0.271
emotion2vec	0.630 ± 0.064	0.248
Mimi	0.616 ± 0.077	0.239
Higgs Audio V2	0.461 ± 0.040	0.236

Table 13: Historical 170k emotion and age results. Emotion variation is across five speaker-disjoint folds.

C Intermediate 170k Checkpoint Evaluation

The updated supervisor report records an evaluation of the 170,000-step intermediate checkpoint from the same training trajectory that later ended at 210,000 steps. It is retained for trajectory provenance but must not be relabeled as a 210k checkpoint result or mixed numerically with Tables 3 and 10. The report predates the current ASR duration-validation path, so its ASR measurements remain diagnostics.

Features	Speaker-ID acc. ↑	Verification EER ↓
CLAE, 170k	0.656	0.280
Random CLAE	0.045	0.394
WavLM	0.296	0.339
Whisper-tiny	0.350	0.375
ECAPA	0.952	0.060
emotion2vec	0.323	0.344
Mimi	0.647	0.231
Higgs Audio V2	0.737	0.185

Table 14: Historical 170k speaker results. Verification is mean+standard-deviation-pooled EER under the all-pairs protocol.

Features	Dev CER ↓	Dev WER ↓
CLAE, 170k	0.826	1.113
WavLM	0.501	0.859

Table 15: Historical 170k fixed-budget attention-ASR diagnostic. The run predates the current duration-validation path.